



Optimal Power Flow

OptimisationOptimal Power Flow

Members

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Project Description :-

This project focuses on solving the Optimal Power Flow (OPF) problem in smart grids to minimize generation costs while maintaining system stability and voltage limits. It involves modeling the power network and applying metaheuristic optimization algorithms such as Genetic Algorithm, Tabu Search, and Simulated Annealing. The study aims to compare the effectiveness of these algorithms on standard IEEE bus systems and explore improvements by integrating renewable energy sources. The project will help enhance the efficiency and reliability of power distribution in modern smart grids.

Goals & Objectives :-

The goal of this project is to minimize generation cost and power losses in smart grids while maintaining system stability and meeting operational constraints. It will achieve this by applying metaheuristic optimization algorithms and comparing their effectiveness on standard grid models.

Objective:-

1. Minimize total generation cost and transmission losses in the electric power grid.
2. Maintain voltage stability and power quality at all grid nodes.
3. Improve integration of renewable energy sources (solar, wind) and distributed generation.
4. Enhance grid reliability, efficiency, and resilience during normal and fault conditions.
5. Optimize both active and reactive power flows for better resource management.

Mathematical Formulation of Optimal Power Flow :-

The Optimal Power Flow (OPF) problem aims to determine the optimal operating state of a power system that minimizes the total generation cost while satisfying all system and operational constraints such as power balance, voltage, and line flow limits.

Objective Function:-

The objective is to minimize the total generation cost

$$\text{Minimize } F = \sum_{i=1}^{N_g} C_i(P_{Gi})$$

F = total generation cost

$C_i(P_{Gi}) = a_i + b_i P_{Gi} + c_i P_{Gi}^2$ is the cost function of generator i

P_{Gi} = real power generated by generator i

N_g = number of generators

Power Balance Constraints:-

total generated power:-

$$\sum_{i=1}^{N_g} P_{Gi} = \sum_{j=1}^{N_d} P_{Dj} + P_{Loss}$$

reactive power balance:-

$$\sum_{i=1}^{N_g} Q_{Gi} = \sum_{j=1}^{N_d} Q_{Dj} + Q_{Loss}$$

where P_{Dj} and Q_{Dj} are the active and reactive power demands at bus j .

Generator Operating Limits

$$P_{Gi}^{min} \leq P_{Gi} \leq P_{Gi}^{max}$$

$$Q_{Gi}^{min} \leq Q_{Gi} \leq Q_{Gi}^{max}$$

Voltage Constraints:-

Bus voltage magnitudes must remain within prescribed limits to ensure system stability.

$$V_i^{min} \leq V_i \leq V_i^{max}$$

Transmission Line Limit:-

$$|S_{ij}| \leq S_{ij}^{max}$$

The apparent power flow in each transmission line should not exceed its thermal limit.

Genetic Algorithm

Genetic Algorithm (GA) is an optimization method inspired by the principles of natural selection and genetics. It works by evolving a population of candidate solutions through selection, crossover, and mutation to find the best solution for complex problems like Optimal Power Flow in smart grids. GA is effective at handling nonlinear constraints and finding near-optimal solutions efficiently in large power systems.

Steps:-

I. Define the Population:

Each individual (chromosome) represents a candidate power flow solution, encoding generator outputs and voltage levels(e.g., $[P_1, P_2, V_1, V_2, \dots]$).

II. Initialize Population:

Start with a randomly generated population of feasible power flow solutions within system constraints.

III. Fitness Function:

Evaluate each solution based on the objective, such as minimizing generation cost and power losses while satisfying voltage and operational constraints.

Hard Constraints:-

1. Generator **active and reactive power limits must not be violated**.
2. Bus voltage magnitudes must stay within lower and upper bounds(e.g. $P_{min} \leq P \leq P_{max}$)
3. **Power balance must** be maintained at each node (**generation = load + losses**).
4. Transmission line flow limits must be respected to prevent overload.

IV. Selection:

Select the fittest solutions for reproduction, giving higher chances to those with better cost and constraint satisfaction.

V. Recombination:

Combine parts of two parent solutions to generate offspring that inherit attributes from both, exploring new regions of the solution space. Swap segments of two parent chromosomes (e.g., switch generator outputs P_1 and P_2) to create offspring solutions.

VI. Mutation:

Introduce small random changes in offspring solutions to maintain diversity and avoid premature convergence.

VII. Termination:

Repeat the process for a fixed number of generations or until the fitness improvement plateaus, indicating convergence on an optimal or near-optimal solution.

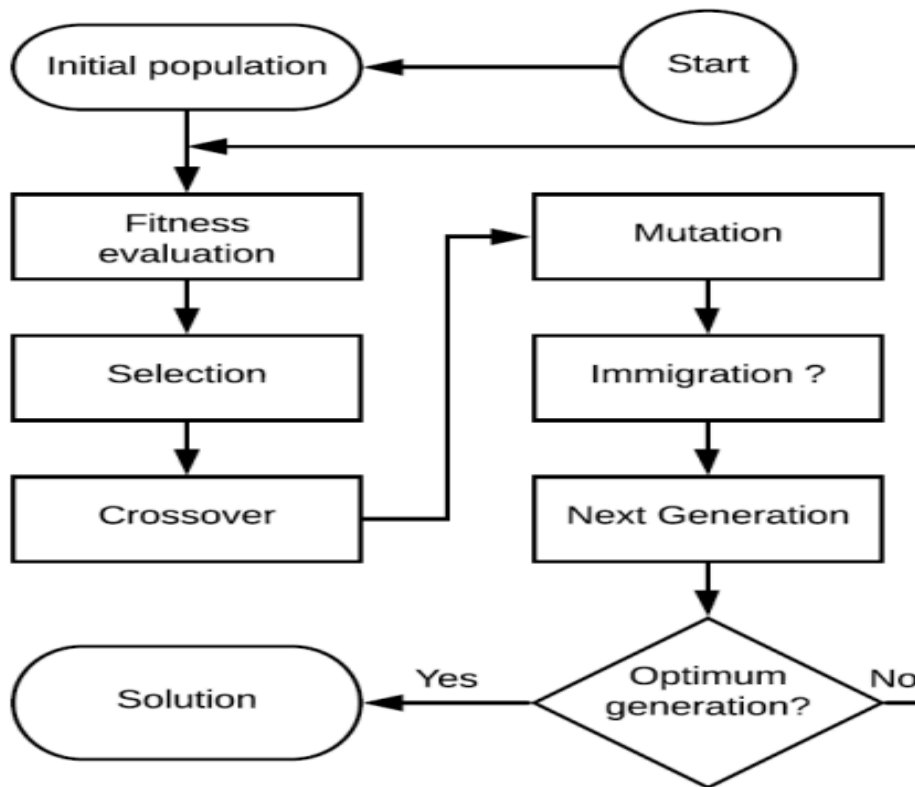


Fig. 1. GA-Based Optimization Approach Flowchart.

Tabu Search

Tabu Search is an **iterative optimization method** that improves a solution by exploring its neighborhood, uses a **memory structure called a tabu list** to avoid revisiting recent solutions, and efficiently escapes local optima to find high-quality or near-optimal solutions in power system problems like OPF(Optimal Power Flow)

The core components of a Tabu Search algorithm are:-

Neighborhood Structure:- Define how to generate neighboring solutions by small changes to the current solution.

Tabu List:- A short-term memory that stores recent moves or solutions to prevent cycling back and encourage exploration.

Aspiration Criteria:- Rules that allow overriding the tabu status if a better solution is found.

Update:- Choose the best non-tabu solution, update the current solution and tabu list accordingly.

Stopping Condition:- Terminate after a fixed number of iterations or when no improvement is found.

Steps:-

I. Initialization:

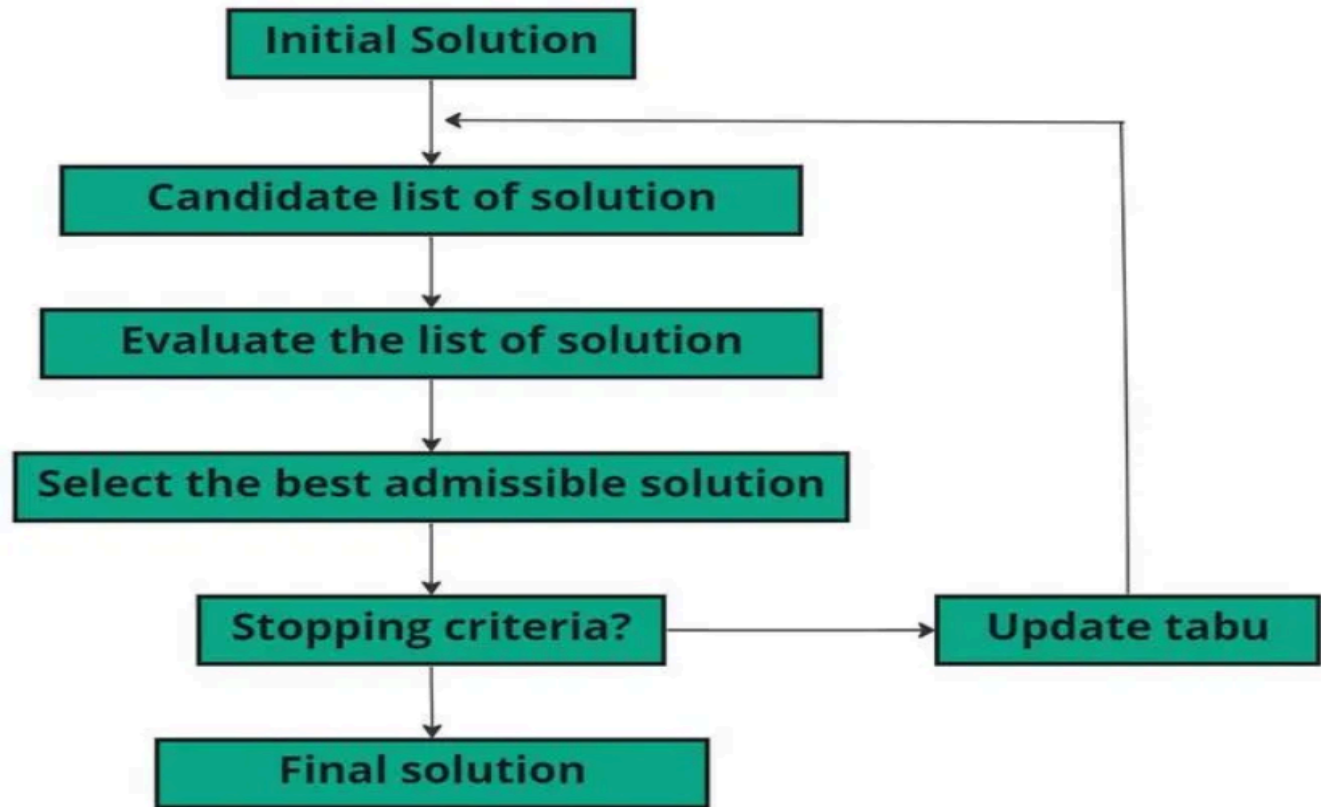
Begin by creating an initial feasible solution, which serves as the **current and best solution initially**. At the same time, an **empty tabu list is created to keep track of recent moves**, and important parameters such as tabu list size (tenure) and maximum iteration count are defined to control the search process.

II. Neighborhood Generation:-

Next, generate a set of neighboring solutions by applying small adjustments or changes to the current solution.

III. Move Evaluation & Selection:-

Then, evaluate the objective function—for example, total generation cost or power loss—of each neighbor. The best neighbor is selected as the next solution.



Flowchart of Tabu Search

Simulated Annealing

Simulated Annealing is a metaheuristic inspired by the annealing process in metallurgy. It explores the solution space by probabilistically accepting not only better solutions but also worse ones early on, with a decreasing probability over time (controlled by "temperature"). This helps avoid local minima and find a near-optimal solution to nonlinear, non-convex problems like OPF.

Key Steps for SA in OPF:-

1. Initialize with a feasible solution (e.g., generator power outputs within limits).
2. Set initial temperature and cooling schedule.
3. Generate neighbor solution by slight perturbation of current generator setpoints or voltages.
4. Calculate objective function (generation cost + penalty for constraint violations).
5. Accept or reject neighbors based on cost difference and current temperature.
6. Cool down temperature.
7. Repeat until stopping criteria (low temperature or no improvement) are met.

Steps:-

I. Initialization

Begin by generating an initial feasible solution for the OPF problem, which serves as both the current and best solution initially. Define the initial temperature, the cooling schedule (how fast the temperature decreases), and set the stopping conditions such as the minimum temperature or maximum number of iterations.

II. Neighborhood Generation

Generate a set of neighboring solutions by applying small, controlled modifications to the current solution. In power flow problems, this could involve adjusting generator outputs or other control variables within permissible limits.

III. Move Evaluation & Acceptance Probability

Evaluate the objective function (e.g., total generation cost or losses) for each neighbor. If a neighboring solution improves the objective function, accept it as the next solution. If it produces a worse result, accept the solution with a probability that depends on the temperature and the magnitude of worsening;

this helps escape local optima. The acceptance probability is typically calculated using the formula:

$$P = \exp(-\Delta/T)$$

where Δ is the increase in objective value and T is the current temperature.

IV. Temperature Update (Cooling Schedule)

After considering a move, decrease the temperature according to the cooling schedule. Common schedules include linear, geometric, or logarithmic cooling. The new temperature is typically set as:

$$T_{\text{new}} = \alpha \cdot T_{\text{old}}$$

with $0 < \alpha < 1$.

V. Best Solution Update

If the newly accepted solution is better than the best solution found so far, update the best solution. Continue to use the current accepted solution (whether better or probabilistically worse) for generating the next set of neighbors.

VI. Stopping Criteria

Repeat the neighborhood, evaluation, acceptance decision, and temperature update steps until the stopping criterion is met—such as reaching the minimum temperature, exceeding the maximum number of iterations, or observing no significant improvement over a set number of iterations. The best solution found is reported as the optimized result.

Test System:-

The **IEEE 14**-bus power system is used for testing and validating the OPF solution. It consists of 14 buses, 5 generators, 11 loads, and 20 transmission lines with specified operating limits. The system provides a standard benchmark for comparing optimization algorithms in power flow studies.



The OPF model is implemented in Python, where generator cost functions, bus limits, and line constraints are applied. The metaheuristic algorithms are tested on the same dataset for performance comparison.

[IEEE 14-Bus System - Illinois Center for a Smarter Electric Grid \(ICSEG\)](http://lamarr.ece.uw.edu/research/pstca/formats/cdf.txt?utm_source=IEEE%2014-Bus%20System%20-%20Illinois%20Center%20for%20a%20Smarter%20Electric%20Grid%20(ICSEG))
[lamarr.ece.uw.edu/research/pstca/formats/cdf.txt?utm_source](http://lamarr.ece.uw.edu/research/pstca/formats/cdf.txt?utm_source=IEEE%2014-Bus%20System%20-%20Illinois%20Center%20for%20a%20Smarter%20Electric%20Grid%20(ICSEG))

Future Scope / Future Improvements

1. Incorporation of Renewable Energy Sources

Solar or wind generation can be added to study the impact of variable power sources on economic dispatch and system stability.

2. Reactive Power Optimization & Voltage Control

The present work focuses mainly on active power. Future work may include reactive power compensation (capacitors, STATCOM, etc.) to improve voltage profile and reduce losses.

3. Security-Constrained Economic Dispatch (N-1 Analysis)

The system can be extended to include contingency analysis, where generator or line outages are considered to ensure reliability and grid security.

4. Multi-Objective Optimization

Instead of only minimizing fuel cost, future models may also consider emission cost, transmission losses, and environmental impact using advanced algorithms like PSO, MOPSO, etc.

5. Time-Series / Real-Time Simulation

The present study uses a single static operating condition. Future work can use hourly load data to perform 24-hour dispatch, integrating demand variation and renewable forecasting.

Final Codes:

 Copy of OPF_Optimization_IEEE14.ipynb

Resources:

Some NP methods

[Optimal Power Flow in Smart Grids: SVC & TCSC](#)

 Lecture 37: Tabu Search

[What is Simulated Annealing - GeeksforGeeks](#)

[Simulated annealing - Cornell University Computational Optimization Open Textbook - Optimization Wiki](#)

Presentation link:

<https://gamma.app/docs/Optimal-Power-Flow-OPF-Project-1efkga6nf1gxohh>